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Exemplification and its local grammar patterns in English as an academic lingua franca in research writing

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ABSTRACT

In this study, we investigate the underexplored rhetorical practice of authors adopting English as a lingua franca (ELF) in academic writing, focusing on their use of exemplification. By giving examples for illustration or clarification, the act of exemplifying is central to the clarity and persuasiveness of research writing. We examined it through a local grammar approach, analysing both its lexico-grammatical and discourse-semantic patterns. Based on the SciELF corpus, which comprises unproofread pre-submission drafts of ELF authors, our analysis shows that *for example* and "Exemplified – Indicator – Exemplification – Subordinate Category" are the dominant marker and local grammar pattern of exemplification. The use of exemplification also relates to the L1 background, professional status and disciplinary alignment of the ELF authors. Czech and Russian authors made the most and least frequent use of exemplification respectively while Finnish and Czech authors often use variant patterns of the dominant one. Additionally, senior academics, particularly from soft disciplines, make the most of exemplificatory markers and prefer the "Exemplified – Indicator – Exemplification – Relevant Studies" pattern. Our study sheds light on the rhetorical language use by ELF authors, and unravels the value of local grammars in the research of English as an academic lingua franca.

1. Introduction

With the increasingly globalised nature of academic communication, participation in the international exchange of information has become an indispensable aspect of scholarly life around the world, and this is especially the case for scholars who are writing in English as a second or additional language. Second-language (L2) speakers are reported to outnumber first-language (L1) speakers by nearly threefold (Mauranen, 2010), and the trend also dominates academic communication (Hyland, 2015; Mauranen, 2012). This points to the issue of using English as an academic lingua franca (academic ELF), which comprises a central concern of academic writing research in recent years (Hyland, 2016; Hyland & Jiang, 2022; McIntosh et al., 2017). From an ESL (English as a second language) and EFL (English as a foreign language) perspective, however, previous studies have typically applied the native norms and rendered English L2 writers disadvantaged by linguistic deficiency (Ingvarsdóttir & Arnbjörnsdóttir, 2013), placing less emphasis on their role as ELF users in writing for research and publication purposes.

Despite the paucity of contributions, Wu et al. (2020) analysed the syntactic complexity ELF writers use in research articles, and turning to lexical bundles, Hyland and Jiang (2022) showed that ELF writers use fewer bundles than those found in published papers.

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Although these syntactic and phraseological uses of academic ELF were suggested, further research on ELF writers is warranted in terms of rhetorical actions and how they are lexico-grammatically realised. Bondi and Borelli (2018), Shchemeleva (2022), and Deng et al. (2023) sought to examine stance-taking practices by ELF writers, but were mainly concerned with metadiscourse resources. To address the research gap, therefore, in this study we set out to explore ELF authors' use of exemplification in research articles, a typical discourse act in academic discourse (Hinkel, 2002; Triki, 2021), and we employ the recently emerging local grammar approach to do so. Simply put, local grammar looks into the lexico-grammatical realisations of one target discourse act and its analysis involves the mapping of meaning elements onto pattern or formal elements (see more in Section 3), making it distinct from traditional grammatical analysis (which typically uses grammatical terms such as *Subject* and *Object* to describe language use in general contexts).

Specifically, this investigation extends local grammar research by applying this approach to explore discourse acts used by EFL writers across different linguacultural backgrounds. We aim to address the following research questions.

(1) What are the forms and frequencies of exemplificatory markers used by EFL writers?
(2) What are the local grammar patterns and distributions of exemplification in EFL writers' texts?
(3) To what extent do the aforementioned exemplificatory markers and local grammar patterns vary across EFL writers' L1 backgrounds, professional status, and disciplinary fields?

2. ELF in professional research writing

Academics around the world have increasingly come under pressure to publish in English-medium international journals, seeing their careers tied to their ability of academic publishing in English (Hyland, 2023; Mur-Dueñas, 2018). It is reported that approximately nine million researchers from 17,000 universities aim to publish in English-language journals annually (Scheegans et al., 2021). Academic publishing, therefore, is now a global industry and principally involves authors, editors, reviewers and readers of different first languages (Mauranen, 2012). All this works to establish the current status of English as the lingua franca of the academic world.

This central role of English in academic publishing, however, is seen to marginalise other academic languages and disadvantage those who do not use English as a first language (Canagarajah, 2002; Flowerdew, 2000). Writing in a second language requires not only a higher level of linguistic proficiency but also an understanding of the rhetorical nuances specific to English academic writing, so often leads to feelings of anxiety and self-doubt among ELF speakers. Therefore, previous studies seek to identify the disparity of discursive practice between English L1 and L2 speakers (Jiang, 2015; Sun & Jiang, 2024) for pedagogical solutions, indicating that writers of other native languages should prioritise an adaptation to Anglophone academic writing conventions. At the same time, however, these conventions themselves are under pressure to change from native varieties of English. With international publishing dependent on the use of one shared language, the swelling number of academic writers from diverse cultural and linguistic backgrounds are poised to reshape this linguistic unity (Mauranen, 2012; McIntosh et al., 2017). Ever more interactions between English L1 and L2 speakers, where expectations are culturally and situationally embedded, create a more complex picture of 'acceptable' academic writing (Hyland & Jiang, 2022).

Accordingly, the "rhetorical agency" (McIntosh et al., 2017, p. 13) of ELF writers in academic prose is found in terms of organising structures (e.g., Lorés-Sanz, 2016) and stance-making expressions (e.g., Xu & Nesi, 2019). However, these studies are mainly based on published texts, which are typically revised and edited for the standard written English of British or American norms (Mauranen, 2012). In addition, while examining unpublished manuscripts, Hyland and Jiang (2022) and Wu et al. (2020) looked to the phraseological and syntactic features of academic writing, while Bondi and Borelli (2018), Shchemeleva (2022), and Deng et al. (2023) focused on metadiscourse resources. However, little attention has been given to rhetorical actions used by ELF writers and the grammatical patterns by which they are expressed. Therefore, there is a clear need to explore the lexico-grammatical realisation of rhetorical actions to unveil "the complex intercultural driving forces of multilingual writers that shape their academic texts", as Connor points out (2018, p. 300).

3. Exemplification and its local grammar in academic writing

Following previous studies, we consider exemplification as a discourse act of elaborating a proposition or illustrating a concept by presenting subsidiary categories or relevant studies, making general arguments specific or abstract concepts concrete (Hyland, 2007; Su & Zhang, 2020; Triki, 2021). It is a recurrent and prominent feature of academic writing (Hinkel, 2004; Hyland, 2007) and plays a crucial role in successful academic writing; in particular, it has been considered "a part of the routine ways in which writers in all fields seek to make their ideas accessible and persuasive" (Hyland, 2007, p. 270) and "a classical and accessible means of rhetorical support" (Hinkel, 2020, p. 408). However, English L2 or novice academic writers are found to have difficulty using exemplification appropriately. For example, Hinkel (2004) reported that "college-level L2 writers rarely employ this strategy [exemplification] successfully" (p.297), and Paquot's (2008) study similarly revealed that the exemplifying phraseologies used by English L2 writers are far more limited than their English L1 counterparts. Furthermore, Su et al. (2022) similarly found that compared with experts, novice writers tend to be less aware of exemplificatory imperatives and make less use of the strategy of exemplifying by citing relevant studies.

Nevertheless, prior studies have typically focused on a limited number of "exemplificatory markers" (Hyland, 2007, p. 270), including *such as*, *for example* and e.g., so they have not fully accounted for the lexico-grammatical realisation of this discourse act.

This lack in the form and meaning mapping is also seen in Paquot's (2008) study, which was restricted to five lexical items realising exemplification (i.e. *for example*, *for instance*, *example*, *illustrate*, and *exemplify*). The narrow analytical lens may obscure a fuller picture of how phraseologies of exemplification are used by English L2 writers in academic texts.

As shown by Su and Zhang (2020) and Su et al. (2021), this weakness can be remedied by a local grammar approach. Briefly, local grammar is a recent development in corpus linguistics, referring to an approach that seeks to account for one meaning or function only (e.g., Cheng & Ching, 2018; Hunston, 2002; Hunston & Su, 2019). In other words, this one-meaning/function-focus makes this grammatical description 'local' or 'specialised'. One advantage of local grammar is that it allows the researcher to explore discourse functions or acts at both lexical-grammatical and discourse-semantic levels; and the defining feature of local grammar is that it uses "transparent and context-specific terminologies", rather than traditional grammatical ones (e.g., Subject, Object), to "analyse corresponding formal elements" (Su & Zhang, 2020, p. 2), thereby contributing to the transparency and specificity of the description. Table 1 exemplifies the local grammar analyses of exemplification.

Table 1 illustrates the mapping of functional elements onto form ones and shows that the realisation of functional terminologies may range from words, phrases to clauses. This kind of analysis can in turn provide a more systematic and specialised description of the target discourse or pragmatic act, as has been shown in multiple studies (e.g., Barnbrook, 2002; Cheng & Ching, 2018; Hunston & Su, 2019; Su & Zhang, 2020; Zhang & Su, 2022). Furthermore, it has also been argued that the resulting local grammar description is informative for the teaching and learning of academic writing, since it may be more pedagogically useful in revealing rhetorical components of the discourse or communicative function than purely formal grammatical descriptions (Hunston, 2002). This points to the significance of local grammars in EAP writing research and pedagogy.

Perhaps more notably, studies have shown that local grammar description facilitates the quantification of the realisational patterns available to perform a given discourse act, thereby contributing to contrastive investigation into discourse acts across contexts (Su et al., 2021; Su et al., 2022). Following this line of research, in this paper we employ the local grammar approach to further explore ELF authors' performance of the discourse act of exemplification in their pre-submission drafts. The investigation into local grammar of exemplification and its possible variations across first language, professional and disciplinary backgrounds can add to our understanding of the influence of contexts on written academic text production, indicating how scholarly authors both use and perceive acceptability. By this study, we also hope to offer some methodological insights for future investigation into ELF authors' performance of other discourse acts in academic writing, thereby enriching the toolkit of (contrastive) academic discourse studies.

4. Corpus and method

4.1. The SciELF corpus

We based the investigation of language use by EFL authors on the SciELF corpus, which was compiled in 2015 at the University of Helsinki and is part of the first publicly available corpus of written academic ELF – WrELFA (http://www.helsinki.fi/englanti/elfa/wrelfa.html). The SciELF corpus comprises original research papers that have not been proofread or edited by language professionals and/or native speakers of English; in other words, the papers are final drafts of unpublished manuscripts (Rowley-Jolivet, 2017). Therefore, it is a corpus of second-language use in written academic communication which "uniquely allows us to study academic texts written by L2 speakers before they are checked by any language brokers" (Shchemeleva, 2022, p. 3).

The SciELF corpus was aimed to curate a diverse range of texts written by authors with different "L1 backgrounds" (Carey, 2015, p. 2), so unpack the variability of L1s to ensure the representativeness of what academic writing in ELF is like (Mauranen, 2012). In this study, we included 146 unproofread pre-submission drafts of research papers from the corpus, which were written by authors with nine different L1 backgrounds across a range of disciplines.[1] The L1 backgrounds of the authors are shown in Table 2.

To explore the influence of contexts on the use of exemplification by ELF authors across different L1 backgrounds, we also included the grouping of research papers into different professional status and broad disciplinary alignment of the authors (Table 3).

4.2. Identification and labelling of exemplification

As shown in previous studies, exemplification is typically signalled by lexical markers (Hyland, 2007; Paquot, 2008; Triki, 2021), indicating that exemplification is amenable to corpus search. Accordingly, we sought to identify a set of the potential items, and used them to search and retrieve instances of exemplification in the SciELF corpus. Drawing on Hyland (2007), Paquot (2008), Su and Zhang (2020) and Triki (2021), we included e.g., *such as*, *for example*, *illustrate*, *for instance*, *an example of*, *exemplify*, and *an instance of* and their variants (e.g., possible variants of *an example of* include *a typical example of, a good example of* etc.) as potential search items. It merits pointing out that, while these items do not certainly comprise an exhaustive list of expressions, searching them, arguably, would warrant a desirable and substantial set of instances of exemplification in any corpus, since they have been shown to be typical exemplificatory markers. Using *AntConc 4.2.4* (Anthony, 2023), all the items were retrieved before we manually analysed each instance from a local grammar perspective, excluding those which do not express the act of exemplification (e.g. *It completes them through giving us an example of Strindberg's fairytale dramaturgy*).

In addition, as we noted above, local grammar uses transparent and context-specific terminologies to analyse instances realising the target discourse act. For our purpose, we followed Su, Zhang and Lu' (2021) set of terms (Table 4) when labelling components of exemplification. These terms were formulated according to principles of transparency and economy, and have been shown to be adequate for accounting for exemplification. Transparency means that the terminologies employed are contextually specific to

[1] There is another Romanian L1 in the SciELF corpus, which has only four texts, so was not included.

Table 1
Local grammar analyses of exemplification[1] (adapted from Su et al., 2021).

Exemplified	**Indicator**	**Exemplification-SC**	
…different kinds of writing tasks	such as	writing the main ideas for different parts of the text	
Exemplified	**Indicator**	**Exemplified-RS**	**Supporting statement**
…has been researched in China for more than ten years.	For example	Liu and Jia (1998)	have used the CD-Rom version of the college English …

[1] All the examples used in the present study, unless otherwise noted, were taken from the SciEFL corpus (see Section 4). The terminologies used for the local grammar analyses of exemplification were taken from Author 2 and others (2021), which were further illustrated in.Table 5 Here, Exemplification-RS stands for "Exemplification-Relevant study", which is typically used to support an argument/observation by citing relevant studies.

Table 2
The nine L1 categories in the corpus.

	First author's L1	no. of articles	no. of words	% of corpus	avg. words per article
1	Finnish	25	123,153	16.78	16.78
2	Czech	22	109,173	14.87	14.87
3	French	16	91,186	12.42	12.42
4	Chinese	21	84,807	11.55	11.55
5	Spanish	13	79,038	10.77	10.77
6	Russian	13	71,376	9.72	9.72
7	Swedish	13	60,060	8.18	8.18
8	Italian	11	58,685	7.99	7.99
9	Portuguese	12	56,625	7.71	7.71
Total		**146**	**734,103**	**100.00**	**5028.10**

Table 3
Professional status and disciplinary alignment of authors in the corpus.[1]

Professional status	**Categories**	**no. of articles**	**no. of words**	**% of corpus**	**avg. words per article**
	Research students	28	157,527	21.46	5625.96
	Junior academics	84	407,212	55.47	4847.76
	Senior academics	34	169,364	23.07	4981.29
Disciplinary alignment	Humanities	23	144,857	19.73	6298.13
	Social sciences	45	255,679	34.83	5681.76
	Life sciences	19	72,881	9.93	3835.84
	Natural sciences	59	260,686	35.51	4418.41

[1] For detailed information on the categorisation of professional status and disciplinary groups, please refer to the documentation of the SciELF corpus (http://www.helsinki.fi/englanti/elfa/wrelfa.html).

exemplification and explicitly reflect the semantic or pragmatic functions of the linguistic elements under analysis. This principle facilitates a clear and unambiguous understanding of the exemplification strategies used by authors. Conversely, economy concerns the use of a limited number of terminologies to avoid unnecessary complexity while ensuring a relatively general and comprehensive description (Su & Zhang, 2020).

Additionally, the distinction between Exemplification-RS (stands for 'Exemplification-Relevant study') and Exemplification-SC (stands for 'Exemplification-Subcategory') is made on the observation that, in academic contexts, examples were used not only to specify abstract superordinate categories by presenting subordinate ones, but also to support an argument/observation by citing relevant studies.

Given the comprehensive discussion of local grammar analyses of exemplification in previous research, in this study we focus more on all the patterns identified in Table 5 (see Section 5). Both authors followed Su, Zhang and Lu' (2021) local grammar analyses of exemplification and achieved a high inter-rater agreement ($\kappa > 0.08$) before resolving disagreements. Then the results were normalised to 10,000 words to allow comparison across the corpora, and *log-likelihood* (*LL*) test was used to determine statistical significances.

5. Overall results of exemplification by ELF writers

We identified 1206 instances of exemplification in the corpus, averaging 16.43 cases per 10,000 words. This frequency is much lower than that in published articles by English L1 scholars, as Hyland (2007), for example, reported 24.5 cases per 10,000 words on average across eight disciplines. This aligns with Paquot's (2008) and Siepmann's (2005) argument that L2 writers' use of exemplificatory expressions is far more limited, suggesting they may not as adept as their English L1 counterparts at glossing over arguments and creating smooth coherence.

Furthermore, 15 local grammar patterns of exemplification were identified. While the analytical procedure was well documented in previous studies (e.g., Su et al., 2021; Su et al., 2022), it is worth pointing out that the current dataset revealed fewer local grammar patterns than the 17 patterns identified in linguistics research articles by English L1 authors (Su & Zhang, 2020). "Hinge +

Exemplification + Indicator + Exemplified" (e.g., See Lillis et al. (2010) for an example of this practice) and "Indicator + Exemplification + Supporting statement" (e.g., as illustrated in examples (6) and (7), a referent can be 'unknown to the speaker' and yet could be 'referring' or 'non-referring') are the two patterns not observed in our corpus, suggesting a distinctive sequence of exemplification used by English L1 authors.

Table 5 shows the 15 local grammar patterns of exemplification in the corpus along with their examples and quantitative frequency. To avoid untypical uses, the subsequent discussion focuses on those patterns that occur more than 15 times in at least one of the corpora.

Although the sequence list of patterns largely concurs with what Su and Zhang (2020) found in research articles by L1 English authors, the Pattern "Exemplified – Indicator – Exemplification-SC" accounted for up to 67.41 % of all the exemplification made by the ELF authors. It is the most explicit and straightforward pattern as to the communicative purpose of providing exemplification. An abstract theoretical or methodological concept is immediately illustrated by the specific examples that follow (see examples 1 and 2). Such a direct pattern seems fairly handy to the ELF authors when they seek to exemplify or illustrate what is inaccessible to readers.

(1) For the purposes of this study, expansions and other accompanying linguistic elements, **such as** adjuncts or circumstances, prove essential for unveiling the way … (Portuguese)
(2) As many new geochronological methods (**e.g.** AMS radiocarbon and cosmogenic exposure age dating) became routine during the last decade …
(Czech)

The patterns "Initiator – Indicator – Exemplification-SC" and "Exemplified - Indicator - Exemplification-RS", together with "Exemplified – Hinge – Indicator – Exemplification-RS" and "Exemplified - Exemplification-SC – Indicator – Supporting statement" were also frequent, comprising approximately 22 % of the total. As a variant of Pattern 1 (Exemplified – Indicator – Exemplification-SC), Pattern "Initiator – Indicator – Exemplification-SC" prepares readers by introducing a subsequent exemplification (3), so it indicates that the ELF authors may feel the need to facilitate readers' reception of the discourse act. Pattern "Exemplified – Indicator – Exemplification-RS" is also a variant of Pattern 1, illustrating an argument or observation by referring to one or more relevant studies (4), but exemplifying by citing relevant studies is substantially less common than by specifying subordinate categories in the corpus, as shown in Table 6 below.

(3) this is the case **for example** of diesel fuel that can be conventional diesel, low sulphur, marketed in Europe. (French)
(4) In Lakoff's view, **for instance** (1987), polysemy is largely natural and characterises any linguistic item. (French)

In contrast, the other five patterns which occurred over 15 times comprise less than 10 % of the total occurrences. This lower frequency might be attributed to their indirect expression of exemplification and greater structural complexity compared to Patterns 1 to 5. Take Pattern 3 (Exemplified – Indicator – Exemplification-RS) and Pattern 9 (Exemplified – Indicator – Exemplification-RS – Supporting statement) for example. Pattern 9 incorporates supporting statements into Pattern 3, demanding a dependence on particular communicative requirements and enhanced skills to navigate the increased grammatical complexity. This added complexity might result in a less frequent use of Pattern 9 compared to Pattern 3. A similar pattern of difference can be also seen between variant expressions 4 and 7, 5 and 8, plus 6 and 10 Table 5.

Table 4
Terminologies for local grammar analyses of exemplification (adapted from Su et al., 2021).

Terminology	Explanation	Example
Exemplified	The proposition or superordinate category that is being illustrated, clarified, or elaborated.	The painter is an example of ***the cosmopolitan artist***.
Exemplification-RS	Elements that specify or illustrate an argument or observation, usually realised by one or more relevant studies.	***Worrall and Zahar (2001)***, for example, argue that the real content of a theory is exhausted by its Ramsey sentence.
Exemplification-SC	Elements that specify or illustrate a more abstract, usually realised by subordinate category/type.	… documents have timestamps of their creation (e.g., ***news articles*** or ***blog posts***).
Indicator	Elements that signal the act of exemplifying.	***For example***, Blank (2007) has compared the production practices …
Initiator	Elements that introduce or initiate the act of exemplifying.	… ***if one does not know***, for example, what is meant by a near-miss …
Researcher	The actor who performs the act of exemplifying.	***We*** will illustrate the use of the strategic quadrant analysis for the assessment …
Hinge	Elements that link different functional elements.	Such simplified or adapted texts ***are*** exemplified in this article by means of Med-E-Pops.
Supporting statement	Elements that serve to further support the illustration of an argument or point.	Putting CALL into practice has been researched in China for more than ten years. For example, Liu and Jia (1998) ***have used to CD-Rom version of the college English textbook*** …

Table 5
Local grammar patterns of exemplification in the corpus.

Pattern	Example						Freq	%
PA 1	**Exemplified**	**Indicator**	**Exemplification-SC**				813	67.41%
	For practical purposes	such as	terminology work or concept analysis …					
PA 2	**Initiator**	**Indicator**	**Exemplification-SC**				102	8.46%
	This is evident,	for example,	in research about the influence of different …					
PA 3	**Exemplified**	**Indicator**	**Exemplification-RS**				74	6.14
	Also according to previous studies	(e.g.	Sherratt, Thornton, & Hatton 2004; Hays & Minichiello 2005 …					
PA 4	**Exemplified**	**Hinge**	**Indicator**	**Exemplification-RS**			46	3.81
	Readerly emotion enjoys much renewed scholarly interest these days	(see	e.g.	Miall 2011).				
PA 5	**Exemplified**	**Exemplification-SC**	**Indicator**	**Supporting statement**			40	3.32
	Other countries,	Sweden	for example,	have been able to criminalise such violations …				
PA 6	**Exemplification-SC**	**Indicator**	**Exemplified**				29	2.40
	The following example	illustrates	the coexistence of subjectivism …					
PA 7	**Exemplified**	**Hinge**	**Indicator**	**Exemplification-SC**			27	2.24
	The internal peculiarities of this type of entrepreneurship	are	exemplified	by re-enactment festivals.				
PA 8	**Exemplified**	**Exemplification-SC**	**Indicator**				24	1.99
	… these patterns	take the form X + marker + Y, as in the phrase smoking causes cancer,	for example.					
PA 9	**Exemplified**	**Indicator**	**Exemplification-RS**	**Supporting statement**			19	1.58
	In addition, some studies arrive at conflicting conclusions.	For instance,	Wright et al. (2005)	find that product innovation does …				
PA 10	**Exemplification-SC**	**Hinge**	**Indicator**	**Exemplified**			17	1.41
	The following	is	an example of	a quiz for "Casablanca".				
PA 11	**Indicator**	**Exemplified**	**Hinge**	**Exemplification-SC**			4	0.33
	An example of	such a realist and pragmatic approach	can be	an understanding that in practice …				
PA 12	**Researcher**	**Indicator**	**Exemplified**	**Exemplification-SC**			4	0.33
	We	will illustrate	this approach	with an excerpt from the corpus composed of resilience and business continuity.				
PA 13	**Exemplified**	**Exemplification-RS**	**Indicator**	**Supporting statement**			3	0.25
	Putting CALL into practice has been researched in China for more than ten years.	For example	Liu and Jia (1998)	have used the CD-Rom version …				
PA 14	**Exemplified**	**Hinge**	**Exemplification-RS**	**Indicator**			3	0.25
	this was one of the reasons why this paper was much cited by other authors	(see	Pan, 2007	for example)				
PA 15	**Researcher**	**Exemplification-SC**	**Hinge**	**Indicator**	**Exemplified**		1	0.08
	I	will use some quantitative compilations	in an effort	to illustrate	how this source reflects Stockholm's shipping trade with Western Europe …			

Table 6
Distribution of exemplificatory markers by L1s in the corpus (per 10,000 words).

Markers	Chinese	Czech	Finnish	French	Italian	Russian	Portuguese	Spanish	Swedish
for example	3.30	1.19	4.63	1.64	2.90	1.54	1.41	2.53	1.67
for instance	0.47	2.47	0.08	2.52	2.90	0.28	0.71	1.01	1.00
an example of	0.35	0.00	0.32	0.33	0.17	0.14	0.53	0.13	0.33
an instance of	0.00	0.00	0.08	0.00	0.00	0.00	0.35	0.00	0.00
such as	3.66	6.41	4.22	2.74	2.22	2.52	5.83	8.73	3.83
e.g.	0.83	6.70	3.33	3.62	4.77	0.14	0.18	0.00	2.66
exemplify	0.35	0.27	0.16	0.00	0.17	0.14	0.00	0.13	0.00
illustrate	0.71	1.19	0.89	0.66	1.02	0.42	1.06	0.63	1.67
Total	**9.67**	**18.23**	**13.71**	**11.51**	**14.15**	**5.18**	**10.07**	**13.16**	**11.16**

6. Results of exemplification by ELF writers across L1 backgrounds

6.1. Frequency of exemplification

While the frequency of exemplification in the corpus is notably lower compared to published texts as aforementioned, a more diverse distribution of markers is found across different L1 backgrounds (see Table 6). Overall, Czech and Russian authors make the most and least use of exemplificatory markers respectively.

More interesting is perhaps the varied distributions of a few similar exemplificatory expressions. Overall, *an instance of* and *exemplify* rarely occurred in the corpora, but *an example of* was strikingly more often used by Portuguese authors, while *illustrate* was relatively most frequently used by Swedish authors. In addition, *for example* was generally often used by the ELF authors, but Czech, French and Italian authors made a relatively greater use of *for instance*. Moreover, *such as* and e.g. were overwhelmingly preferred by Spanish and Czech authors respectively. This varied use of exemplificatory markers across L1 backgrounds expands upon the findings of Guziurová (2020), who investigated the usage among ELF writers in general. Beyond the lexical markers, it is more informative to examine the patterns of exemplification used by the ELF authors from the local grammar perspective.

6.2. Patterns of exemplification by writers of different L1 backgrounds

Similar to the distribution of exemplification markers, the local grammar patterns were used in dissimilar ways by authors from different L1 backgrounds. Table 7 presents the varied distribution of the patterns used by different L1 authors in the corpus.

Pattern 1 (Exemplified – Indicator – Exemplification-SC) seems equivalently preferred by all the ELF authors, while registering the highest and lowest frequency in the manuscripts of Czech and Russian authors respectively. However, its variant patterns 2 (Initiator – Indicator – Exemplification-SC) and 3 (Exemplified – Indicator – Exemplification-RS) were more frequently used by Finnish and Czech authors, but notably none of these patterns were found in Russian authors' texts. In addition, it is interesting that as a variant of Pattern 3, Pattern 9 (Exemplified – Indicator – Exemplification-RS – Supporting statement) was much more often used by Chinese authors while seeing a low frequency of use by the European counterparts. Furthermore, Pattern 4 (Exemplified – Hinge – Indicator – Exemplification-RS) showed a frequent use by Italian authors, but its variant Pattern 7 (Exemplified – Hinge – Indicator – Exemplification-SC) was more often employed by Russian than other ELF authors.

Additionally, Patterns 5 (Exemplified – Exemplification-SC – Indicator – Supporting statement) and 8 (Exemplified – Exemplification-SC – Indicator) were highly preferred options to Italian and French authors respectively, although the two variants only differ by the supporting statements added to the exemplification. Pattern 6 (Exemplification-SC – Indicator – Exemplified) is strikingly preferred by Swedish authors and sees a low frequency in other ELF texts. As its variant, Pattern 10 (Exemplification-SC – Hinge – Indicator – Exemplified), which includes a linking element between the exemplar and what is being exemplified, was frequently used by Chinese, Portuguese and Finnish authors, although its frequency was relatively low in the corpus.

Furthermore, the peculiarity of L1 backgrounds of adding supporting statements is also notable, particularly among Italian and Chinese authors, albeit for different reasons. As illustrated in examples (5) and (6), Italian authors typically prefer to expand on a

Table 7
Distribution of exemplifying patterns by L1s in the corpus (per 10,000 words).

Patterns	Chinese	Czech	Finnish	French	Italian	Russian	Portuguese	Spanish	Swedish
PA 1	6.60	11.56	8.11	7.68	7.16	3.58	6.93	11.18	7.33
PA 2	0.59	1.28	1.71	0.00	1.19	0.00	0.53	1.27	0.67
PA 3	0.55	2.67	0.73	0.66	0.85	0.00	0.58	0.00	0.50
PA 4	0.00	1.19	0.73	0.33	2.59	0.14	0.00	0.00	0.17
PA 5	0.00	0.18	0.57	0.66	1.36	0.14	0.53	0.00	0.50
PA 6	0.27	0.18	0.08	0.11	0.00	0.00	0.31	0.00	1.40
PA 7	0.12	0.37	0.24	0.22	0.34	0.92	0.35	0.13	0.00
PA 8	0.00	0.18	0.41	1.41	0.00	0.00	0.00	0.00	0.00
PA 9	0.79	0.18	0.32	0.00	0.17	0.00	0.00	0.00	0.00
PA10	0.35	0.00	0.32	0.00	0.00	0.00	0.35	0.13	0.17

subordinate category, whereas Chinese authors focus more on elucidating relevant studies in their construction of exemplifications.

(5) In the case of generalised credit bubbles, **for example**, the key instruments are: anticyclical capital buffers and liquidity requirements.

(Italian)

(6) Putting CALL into practice has been researched in China for more than ten years. **For example**, Liu and Jia (1998) have used the CD-Rom version of the College English textbook to teach their undergraduate students in a university in Guangzhou. (Chinese)

The above cross-cultural variation can be attributed to a complex interplay of linguistic, cultural, and educational factors. The structural differences between the writers' L1s and English may influence their preferences for certain patterns, as they may transfer syntactic structures or rhetorical strategies from their native languages (Wu et al., 2020). For instance, the preference of Swedish authors for Pattern 6 might reflect a common structure in Swedish academic writing. Cultural rhetorical traditions also play a crucial role in shaping writing styles. The tendency of Italian and Chinese authors to add supporting statements may stem from culture-specific expectations of elaboration and explicitness in their discourse (Yu, 2021). Furthermore, variations in educational practices and the level of exposure to English-language academic discourse across countries can lead to different conventions in scholarly writing. For example, the focus of Chinese authors on elucidating relevant studies might reflect a stronger emphasis on literature review in their academic instruction.

7. Results of exemplification by ELF writers across professional status

7.1. Frequency of exemplification

Another important contextual aspect underlying writers' language choices is the experience of writing and publishing they bring to the act. We can see from Table 8 that senior academics made the most use of exemplificatory markers among the author groups ($LL = 11.33$, $p < 0.001$), clarifying claims through illustrative material and then creating a plausible argument. While exemplification is a routine way of exposition writers use to make their ideas accessible (Rodríguez-Abruñeiras, 2017), senior academics most commonly availed themselves of the rhetorical means and exhibited certain ability to identify where readers may need help in interpreting points and where elaboration and examples were needed.

More specifically, senior authors made a significantly greater use of *such as* and *for instance* than research students and junior academics ($LL = 20.18$, $p < 0.001$; $LL = 33.67$, $p < 0.001$). The marker *such as* introduces instances exemplifying or illustrating a general designation, and allows writers flexible leeway of managing the number of exemplar items (7 & 8).

(7) This exhibition, directed by Kurt W. Forster, with the theme of Metamorph, included spectacular architecture by leading architects **such as** Alsop, Eisenman, Libeskind, Gehry, Hadid, Ito, Koolhaas, Miralles, Siza, Navarro Baldeweg. (Senior academics)

(8) With time-series data from Netherlands 1975–2005, Ingen and Dekker (2011) have examined whether societal developments **such as** educational expansion, secularisation, and changes on the job market affect levels of volunteering in the Netherlands. (Senior academics)

Compared with the other two groups, senior academics seem apt to employ the rhetorical expediency *such as* offers, and also frequently use *for instance*, which typically opens a new sentence and is able to link various syntactic forms (9 & 10). However, research students tended to use *for example* to achieve such communicative ends (11 & 12) since *for example* and *for instance* function in a similar way (Rodríguez-Abruñeiras, 2020) and are the prototypical lexico-grammatical resources that are taught in classroom (Paquot, 2008).

(9) The skipper, Anders Dahlstrom, **for instance**, passed through the sound 18 times during the seven studied year. (Senior academics)

Table 8
Distribution of exemplificatory markers by professional status in the corpus (per 10,000 words).

Markers	Research students	Junior academics	Senior academics
for example	4.16	2.02	2.03
for instance	1.05	1.17	2.09
an example of	0.38	0.27	0.20
an instance of	0.10	0.02	0.00
such as	4.60	3.33	6.08
e.g.	0.80	1.77	1.75
exemplify	0.04	0.80	0.15
illustrate	0.41	1.11	0.62
Total	**11.54**	**10.49**	**12.92**

(10) **For instance**, while uncontrolled anger is as basic emotion marked by particular glottal behaviour, violent articulatory movement and extreme articulatory position, when, it conveys a conscious negative predisposition, it is first of all marked by prosodic features.

(Senior academics)

(11) **For example**, people are acquainted with somebody who defaulted strategically are more likely to declare their intention to do so.

(Research students)

(12) Turkka criticised **for example** the educational system in which the students have to learn by sitting and staying quiet in place (tv-document 1994). (Research students)

To some extent, junior academics stood out in their more frequent use of e.g. and *illustrate* than the other author groups, mainly offering either representative references (13) or visual material (14) in support of a given standpoint.

(13) The second group represents numerous studies, where main arguments for palaeoseismic origin of landslides are their extraordinary size, mechanism and location near active faults (**e.g.** fault-bounded marginal slope of mountains; Balescu et al., 2007). Many dated rock avalanches situated along the Alpine fault (New Zealand) probably originated due to the major earthquakes (**e.g.** Chevalier et al., 2009; Dufresne et al., 2010; Barth, 2014). (Junior academics)
(14) As Table 2 **illustrates**, variables of bed gradient and amount of LWD channel indicate strong correlation and they are grouped in to principal component 1. (Junior academics)

The difference is more striking when we turn to the local grammar patterns of exemplification used by the different authors across professional status.

7.2. Patterns of exemplification by writers of different professional status

As shown in Table 9, Pattern 1 (Exemplified – Indicator – Exemplification-SC) dominates the distribution across professional status as much as across L1 backgrounds of authors as aforementioned. This pattern foregrounds what is exemplified and then immediately elucidates propositions by specific cases (Su & Zhang, 2020), so offers writers the most accessible and easiest way to make the abstract more concrete and develop argumentation (15, 16 & 17).

(15) Meanwhile, the method relying on ground station has something the matter, **such as** time delay and lower navigation precision.

(Research students)

(16) As for the relative dating, most approaches are based on rock weathering (**e.g.** weathering rinds, Schmidt Hammer rebound values or rock varnish microlaminations; Whitehouse, 1983; Orwin, 1998; Clark and Wilson, 2004; Douglass et al., 2005; Mills, 2005). (Junior academics)
(17) There may be many ways to do it. **For instance**: i) promoting the voluntary option for a type of marriage characterised legally by a greater stability … (Senior academics)

In addition, Pattern 2 (Initiator – Indicator – Exemplification-SC) was relatively often used by research students and junior academics, while Patterns 7 (Exemplified – Hinge – Indicator – Exemplification-SC) was more frequent in senior academics' texts. Pattern 2 is the most direct variant of Pattern 1, constructing exemplification by subordinate material in a similar palpable way to Pattern 1 (Su

Table 9
Distribution of exemplifying patterns by professional status in the corpus (per 10,000 words).

Patterns	Research students	Junior academics	Senior academics
PA 1	9.59	7.68	7.56
PA 2	1.14	1.01	0.30
PA 3	0.19	1.08	0.77
PA 4	0.76	0.69	0.12
PA 5	0.51	0.44	0.24
PA 6	0.13	0.27	0.18
PA 7	0.13	0.10	0.94
PA 8	0.13	0.34	0.24
PA 9	0.06	0.05	0.65
PA 10	0.06	0.22	0.06

et al., 2021), so emerges as a preferred option among research students and junior academics. The use of Pattern 2, however, is relatively restricted to the connection of exemplification with prior discourse (18 & 19).

(18) Furthermore, habitus appears **for example** in body size, shape, postures and gestures, in the volume and tone of one's voice, in the action and facial expressions, in ways of speaking and taking space and time, and in the ways of relating to others (cf. Bourdieu 2000:138, 1990).

(Research students)

(19) Upon further reflection, **e.g.** during class discussion, it can always turn out you were a poor (mind-)reader and misconstrued much of a literary character's motives and emotions. (Research students)

While as a variant of Pattern 1, Pattern 3 comprises a dual rhetorical presentation of material, exemplifying an abstract and complicated string of thoughts and situating a new claim in the scaffolding of already accredited facts by academic citation (20 & 21). Thus both junior and senior academics were inclined to employ this pattern to create a rhetorical context of clarification and attribution of academic knowledge.

(20) The XSH Fm. generally consists of light brown to yellow mudstones, with intercalated sandy and conglomeratic lenses that contain many invertebrate and mammal fossils (**e.g.**, Qiu et al., 2001; Wang et al., 2005), fossilised plants (**e.g.**, Geng et al., 2001; Sun et al., 2004), and sporomorphs (Li, 2000). (Junior academics)
(21) These points can be supported also by drawing on world system-theory methodologists **such as** McMichael (1992), emphasising that comparisons should be incorporated comparisons … (Senior academics)

Furthermore, it is also observed that Pattern 7 (Exemplified – Hinge – Indicator – Exemplification-SC) and 9 (Exemplified – Indicator – Exemplification-RS – Supporting statement) were often used by senior academics in their research writing. Pattern 9 adds supporting statement to Pattern 3 and strengthens reasoned coherence as in (22), while Pattern 7 allows writers to elaborate on illustrative material and develop argumentation (23). Both may suggest certain familiarity with disciplinary machinery of academic persuasion.

(22) In many Western European nation states Protestantism, Lutheranism in the North, was seen to be the people's religion as opposed to the Roman Catholic Church and the aristocracy in the sixteenth and seventeenth century. **For example**, Aage Sorensen (1998) has shown that the Kings of Denmark-Norway joined forces with Lutheran Pietism since 1660 until the mid-nineteenth century. (Senior academics)
(23) In West Central Africa, this was well **illustrated** by Sergeant-major Afonso Mendes, who around 1650 wrote one of the most remarkable works on seventeenth century medicine in Africa. (Senior academics)

We may see a combination of factors related to writing experience, rhetorical sophistication, and familiarity with disciplinary conventions at play. Senior academics' more frequent use of exemplificatory markers suggests a heightened awareness of reader needs and a refined ability to anticipate where elaboration is necessary. This rhetorical investment likely stems from their prolonged engagement in academic discourse. Research students' preference for the use of *for example* may reflect their reliance on more explicitly taught rhetorical devices, indicating a stage of academic writing development where they adhere closely to learned patterns. Junior academics' rhetorical choices, on the other hand, appear to represent a transitional phase, where they are expanding their repertoire of academic expressions towards the nuanced approach of senior academics.

The distribution of more complex patterns, such as those incorporating relevant studies or supporting statements, among junior and senior academics, reflects a growing sophistication in argumentation and a deeper understanding of disciplinary expectations for evidence and elaboration. This progression from simpler to more complex exemplification strategies across academic status illustrates the gradual acquisition of research writing norms and the development of a more nuanced rhetorical awareness that comes with increased enculturation into academic discourse communities (Prior & Bilbro, 2012).

Table 10
Distribution of exemplificatory markers by disciplines in the corpus (per 10,000 words).

Markers	Humanities	Social sciences	Life sciences	Natural sciences
for example	2.72	4.15	0.55	1.84
for instance	2.61	1.33	0.11	0.94
an example of	0.56	0.24	0.00	0.08
an instance of	0.11	0.03	0.00	0.00
such as	6.88	7.26	1.88	4.49
e.g.	4.11	4.89	5.40	7.55
exemplify	0.17	0.24	0.00	0.00
illustrate	2.28	0.80	0.22	0.78
Total	**19.42**	**18.93**	**8.16**	**15.68**

8. Results of exemplification by ELF writers across disciplines

8.1. Frequency of exemplification

ELF writers' use of exemplification is also situated in a disciplinary context. As shown in Table 10, exemplification occurred more frequently in the texts of humanities and social sciences than those in life and natural sciences ($LL = 53.62$, $p < 0.001$).

This difference provides evidential support to variations in the way other metadiscoursal elements are used to mediate disciplinary contexts (e.g. Hyland, 2007; Jiang, 2017; Jiang & Hyland, 2021). Unlike hard sciences, which see knowledge as developed through relatively cumulative and tightly structured growth (Becher & Trowler, 2001), humanities and social sciences do hardly share a specialised and steady rhetorical context of investigation and resulting claims (Hyland & Jiang, 2019), and a typically less cohesive readership requires a restructuring of more diverse components of what is discussed (Hyland, 2007). Thus exemplification plays a role in encouraging readers to recognise a phenomenon through familiar references or recoverable experiences and become involved in the unfolding text, pointing to a known and recoverable reality (24 & 25). In other words, examples in soft knowledge fields mark a heavy interactive investment in contextualisation, seeking to persuade the reader of abstractness through the concreteness which actually exists (Su et al., 2021).

(24) The complexity of the issue can be **illustrated** by the study of Ren et al. (2008: 8), who propose a framework to model causal relationships on offshore safety assessment focusing on human and organisational factors.

(Humanities)

(25) Scientific uncertainty arises when the mechanisms of actual emission processes are not completely understood. **For example**, a lot of direct and indirect factors associated with global warming potential involve significant scientific uncertainty. (Social Sciences)

However, it is interesting to find that writers in the hard domains relied much on e.g. while *such as* was preferred in soft disciplines. As seen in (26), e.g. enables exemplification to be separated from the remainder of the proposition by brackets, allowing writers to indicate that the enclosure provides illustrative material rather than main content, and this typically characterises scientific writing (Hyland, 2007). Writers in the soft fields, by contrast, were more inclined to use *such as* to interpolate a practical and elaborate illustration in the on-going text to support a prior locution (27).

(26) there is a remarkable trend in using cosmic ray exposure techniques for dating of both accumulation (**e.g.** landslide boulders) and depletion (**e.g.** landslide scarps) parts of landslides. (Natural Sciences)

(27) The reasons for their settlement also involved the Russian conquest of the Caucasus, and its economic and social consequences, **such as** the radicalization of the northern Caucasus under the rule of Imam Shamil, as well as the highlanders' customary law of adat (blood feud) practised in Chechnya. (Humanities)

We also observed a notably rare presence of *an example of* and *exemplify* in hard sciences although they are mundane ways of exemplification, while *an instance of* appears to be a less favoured choice among ELF authors in all fields. We now turn to local grammar patterns of exemplification in disciplinary writing of ELF authors.

8.2. Patterns of exemplification across disciplines

As we have seen above, exemplification represents disciplinary practices, construing the "characteristic structures of knowledge domains and argument forms particular to the disciplines that create them" (Hyland, 1999, p. 109), and this is also noted in local grammar patterns of exemplification (Table 11).

Pattern 1 (Exemplified – Indicator – Exemplification-SC) is most common across disciplines, although its difference from Pattern 2

Table 11
Distribution of exemplifying patterns by disciplines in the corpus (per 10,000 words).

Patterns	Humanities	Social sciences	Life sciences	Natural sciences
PA 1	11.08	12.04	6.59	8.55
PA 2	0.52	0.71	0.13	3.52
PA 3	1.11	1.77	0.4	0.88
PA 4	1.31	0.55	0.13	0.46
PA 5	0.87	0.28	0.26	0.34
PA 6	0.85	0.45	0.13	0.42
PA 7	1.44	0.49	0.13	0.11
PA 8	0.62	0.28	0	0.19
PA 9	0.29	0.24	0	0.23
PA 10	0.13	0.42	0	0.11

is remarkable in soft knowledge fields. Moreover, Pattern 3 (Exemplified – Indicator – Exemplification-RS) is more frequent in academic prose in the soft than hard knowledge fields, constructing explanations for research subjects in social worlds by relevant studies (28 & 29).

(28) susceptibility to mental imagery has long been known to enhance text comprehension and learning in children (**e.g.**, Gambrell and Bales 1986; Anderson and Kulhavy 1972). (Social sciences)
(29) Knowledge patterns have been studied by a number of researchers, mostly in English (**e.g.** Ahmad & Fulford 1992; Christensen 2000; Feliu 2004; Meyer 2001; Pearson 1998). (Humanities)

Additionally, among the two, Pattern 1 registered a higher proportion in soft disciplines, which indicates a predilection for elaborating and contextualising a statement by citing a particular subordinate in order to represent the category of events, people or phenomena mentioned in the immediately prior text (30 & 31). Thus by using such a pattern of exemplification, writers help readers to understand the target concept by relating the unfamiliar to the more recognisable through a specific case, and weave illustrative stories or scenarios into their academic discourse which are deeply rooted in cultural and social contexts.

(30) By applying reflexive body techniques, **for example** making and dressing up, one puts herself in relation with the objectified image of herself. (Humanities)
(31) Marketing variables include qualitative and quantitative aspects, **such as** new market shares and customer satisfaction (this last example is dedicated more to product's Level C than to the project's Level B).

(Social sciences)

In addition, humanities made a much more frequent use of patterns 4 (Exemplified – Hinge Indicator – Exemplification-RS) and 7 (Exemplified – Hinge – Indicator – Exemplification-SC) than the other fields ($LL = 13.97$, $p < 0.001$; $LL = 10.61$, $p < 0.001$), which share a core component "Exemplified – Hinge – Indicator – Exemplification". As seen in examples (32) and (33), the elements that link different parts are typically what Hyland (2017) calls "textual acts", which direct readers to another part of the text or to another text. The heavy presence of such a pattern indicates that in humanities "the research is less linear, the literature more dispersed and the readership more heterogeneous" (Hyland, 2017, p. 10). Therefore, writers cannot presuppose a shared context to the same extent, but have to build one far more through extensive textual referencing.

(32) Call Conformists the views that subscribe to the former (see **e.g.** Christensen 2007; Elga 2007; Feldman 2006), and Non-Conformists the views that subscribe to the latter (see **e.g.** Conee 2010; Wedgwood 2010).

(Humanities)

(33) Nevertheless, it would make sense only if LOP followed this lexical pattern throughout her translation, which has not happened (see, **for instance**, the lexical choice *Lorde Henry*, in excerpt 5). (Humanities)

It is also notable in Table 11 that Pattern 2 (Initiator – Indicator – Exemplification-SC) was far more frequently used in natural sciences ($LL = 69.23$, $p < 0.001$). Grammatical elements are used, such as prepositional or verbal items, to introduce or initiate exemplifying content, and the writers' rhetorical choices are a result of the interactions of disciplinary members engaged in a common pursuit. The following two typical extracts may give us some taste of how exemplification is used to construe the textual interaction.

(34) The REY concentrations measured in our study are higher than values previously reported for speleothems, with **for example**, La concentrations between 115 and 2835 ppb in this study. (Natural sciences)
(35) More efficient dispersal in marine systems due to the lack of dispersal barriers enhance SS over neutral dynamics in oceans compared **e.g.** with lakes or terrestrial systems where neutral forces may be stronger due to a more restricted dispersal. (Natural sciences)

The stark variations in exemplificatory practice between soft and hard disciplines reflect the different ways knowledge is constructed and communicated. In soft disciplines, the more frequent use of exemplification, particularly through patterns that elaborate on statements with subordinate categories or relevant studies, relates to the interpretive and context-dependent nature of knowledge in these fields. This approach helps writers navigate a more heterogeneous readership, ground abstract concepts in concrete scenarios, and strengthen rhetorical persuasion. The preference for *such as* and the heavy use of "textual acts" in humanities further illustrate the need to guide readers through complex, interconnected ideas in a non-linear fashion, reflecting the dispersed nature of literature and the cultural sensitivity required in these fields (Becher & Trowler, 2001).

In contrast, hard sciences demonstrate a different approach to exemplification, with a notable preference for e.g. and more structured patterns of presenting examples, such as Pattern 2 with its initiating elements. This reflects the cumulative and tightly structured nature of knowledge in these disciplines, where a more specialised audience shares a common background, and empirical evidence often takes precedence over illustrative examples (Hyland, 2015). The lower frequency of exemplification in hard sciences aligns with the more linear argumentative structure and the emphasis on clear, replicable methods and results. These contrasting patterns underscore how the epistemic cultures of different disciplines shape research writing practice, influencing how academics use

exemplification to construct arguments and engage readers.

9. Conclusions

This study has explored the discursive practices of ELF authors in international publishing, which is a growing issue for EAP researchers and practitioners. In particular, we have sought to understand the ways exemplification was performed by writers with different L1 backgrounds, professional status and across different disciplines. We examined the form, frequency and local grammar patterns of exemplification in research writing and considered contextual factors that have possibly influenced the use of exemplification. Our study found that, overall, *for example* and "Exemplified – Indicator – Exemplification-SC" are the most frequent marker and local grammar pattern of exemplification among ELF authors, indicating their prevalence in using English as an academic lingua franca for this rhetorical action.

Additionally, the choice of exemplifying markers and local grammar patterns by ELF authors are impacted not only by the L1 but also by the cultural backgrounds, probably the national academic traditions. Notably, Czech and Russian authors made the most and least use of exemplification respectively despite both languages belonging to the Slavic language family. Despite the wide prevalence of Pattern 1 (Exemplified – Indicator – Exemplification-SC), its variant patterns 2 (Initiator – Indicator – Exemplification-SC) and 3 (Exemplified – Indicator – Exemplification-RS) were more frequently used by Finnish and Czech authors, but were absent in Russian authors' texts. Furthermore, professional status and discipline also present an important contextual variable in ELF writing. Senior academics, particularly from soft disciplines, made the most of exemplificatory markers. Distinct preferences for local grammar patterns were observed between senior and junior academics, with the former favouring Pattern 3 and the latter Pattern 7 (Exemplified – Hinge – Indicator – Exemplification-SC), differing from research students who predominantly used Pattern 2. This suggests that more experienced ELF authors have available to them more flexible or complex resources for performing exemplification. Regarding disciplinary variations, Pattern 3 was more prevalent in the humanities and social sciences, Pattern 2 more common in the natural sciences, although Pattern 1 still dominates across the disciplinary spectrum.

We must emphasise, in agreement with Hyland and Jiang (2022), that our research is descriptive, aiming to illustrate how ELF authors typically write, rather than prescribing how they should write. Consequently, the rhetorical language choices made in ELF authors' texts should not be considered 'incorrect', 'unsuitable', or 'non-native' in a way that undermines the agency and legitimacy of the ELF writing community in the international academic arena. Importantly, every additional paper authored by an ELF writer enriches the evolving landscape of academic English, which is continuously shaped and reshaped through its usage. Academic discourse is a vibrant blend of diverse English forms, continuously influenced by the growing involvement of ELF authors in international publishing. This involvement is likely to gradually expand not only the range of metadiscourse resources, syntactic and phraseological features, but also local grammar patterns of discourse acts to be observed in professional writings.

Moreover, local grammars add to our current knowledge of phraseology of linguistic choices by a special emphasis on the discourse-semantic patternings, which in turn points to the need to employ the local grammar approach to further explore ELF writers' performance of other discourse acts such as definition, summarisation, indicating limitations etc. Such investigation holds both theoretical and pedagogical value in the research and teaching of English as an academic lingua franca. In particular, local grammars offer insights into the patterned nature of academic language produced by ELF writers in today's scientific communication, not only in terms of lexical-grammatical forms, but also discourse-semantic patternings. In the meantime, they can also help to furnish ELF writers with an inventory both of lexical-grammatical forms and their associated discourse-semantic patterns, thereby enriching their linguistic choices for meaning-making in academic contexts. Seen in these lights, local grammars can be considered a useful methodological candidate to further investigation into ELF academic writing, in particular ELF writers' performance of discourse acts, which could also have important pedagogical implications for academic writing teaching in ELF context.

The present study is of course not without limitations. One limitation relates to the identification of exemplification in the corpus. We basically used a marker approach to identify instances of exemplification, leaving inevitably those exemplification cases undetected which do not co-occur with the predetermined exemplificatory markers. Another limitation is that the corpus we used is relatively small, which may raise issues concerning the representativeness of the data and consequently the generalisability of the findings reported in the present study. Nevertheless, we hope our study have offered some insights into ELF academic writers' use of exemplification and, more importantly, provided a viable methodological approach via which ELF academic writers' performance of other discourse acts can be investigated.

CRediT authorship contribution statement

Feng (Kevin) Jiang: Writing – original draft, Validation, Investigation, Conceptualization. **Hang Su:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis.

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